

OpenStack Fundamentals

Part 1 of 2

*Private Cloud Architecture
& Core Services*

ADSAI – HCP and Cloud Course

University of Trieste

Academic Year 2025/2026

Topics: Why Private Cloud • OpenStack Architecture • Core Services • Keystone •
CINECA ADA

Learning Objectives

What you will be able to do by the end of Part 1

Explain

why private cloud remains essential in scientific computing (GDPR, cost, HPC integration)

Configure

CLI authentication to OpenStack environments, including CINECA ADA Cloud

Describe

the modular OpenStack architecture and the role of each core service

Deploy

VMs, configure networking, security groups, and attach block storage volumes

Map

OpenStack services to AWS equivalents, identifying architectural similarities and differences

Differentiate

the responsibility model: managed public cloud (AWS) vs. private cloud (OpenStack)

The Cloud Deployment Spectrum

From fully public to fully private — where does OpenStack fit?



Private Cloud: On-demand self-service, resource pooling, rapid elasticity, measured service (NIST SP 800-145) applied to **infrastructure the institution owns or directly controls**. The critical distinction: who operates the platform and where data resides.



Not just 'old-fashioned on-premises'

Same cloud principles (NIST 800-145) applied to self-owned infrastructure



Same user experience, different operator

Self-service portal, API-driven provisioning: users don't see the difference



Data sovereignty and control

Data never leaves your datacenter; full audit trail; configurable hardware

Why Private Cloud? Key Drivers in Scientific Computing

Six reasons research institutions choose private over public cloud



Data Sovereignty

GDPR, national regulations prohibit cross-border transfer of personal research data (genomics, clinical, surveys). EAR export controls on dual-use research.



TCO at Scale

Large, stable compute demand (>70% utilization): CapEx model over 5-10 years beats on-demand or reserved pricing. Scientific computing clusters running continuously.



Hardware Customization

Custom InfiniBand topologies, GPU direct interconnects, specialized storage (Lustre, GPFS). Public cloud cannot expose these configurations.



HPC Integration

LAN-speed access to parallel filesystems. Moving hundreds of TBs over WAN to public cloud is a bottleneck: colocation eliminates it.



Regulatory Compliance

Shared responsibility model complicates audits. Full stack control: you own the audit trail, HSM, access logs. Medical, defense, ESA data.



Network Topology

Custom RDMA fabrics, SR-IOV direct device assignment, high-radix fat-tree topologies impossible to replicate in public cloud VPC constraints.

When NOT to Choose Private Cloud

Honest trade-off analysis....the hybrid model is often optimal

✓ Choose Private Cloud when...

- Stable, high-utilization workloads (>60-70%)
- Sensitive data subject to GDPR / export controls
- Custom hardware topology required (InfiniBand, RDMA)
- Strong in-house platform operations team
- Long-term budget committed to infrastructure CapEx
- Need for full audit trail and data locality

✗ Prefer Public Cloud when...

- Bursty or unpredictable workload patterns
- Small team, no platform operations expertise
- Need managed services (serverless, ML endpoints)
- Geographic distribution or global availability
- Short project timeline (no CapEx justification)
- Rapid prototyping: avoid infrastructure setup time

Optimal Strategy Hybrid model: private infrastructure for stable, sensitive baseline workloads; public cloud burst for peaks. CINECA ADA Cloud + AWS is exactly this pattern.

OpenStack: Origins and Governance

From NASA/Rackspace joint project to global open-source standard

2010	NASA (Nebula) + Rackspace (Cloud Files) joint release. Modular design from day 1.
2012	OpenStack Foundation formed. >150 corporate members. 6-month release cycle begins.
2014	Juno release: mature compute, networking, storage. Wide datacenter adoption begins.
2020	Renamed Open Infrastructure Foundation (OIF). Scope expands beyond OpenStack.
2021	Wallaby (2021.1): version used at CINECA ADA Cloud.
2024	Dalmatian (2024.2): current stable release. Continuous feature development.

"OpenStack controls large pools of compute, storage, and networking resources throughout a datacenter, managed via dashboard or API." Open Infrastructure Foundation

OpenStack in Scientific Computing

Dominant platform for national and international research infrastructure

CERN

Largest OpenStack
deployment: 10K+ cores

EOSC

European Open
Science Cloud

GÉANT

National Research
Networks across EU

6 months

Release cycle
Wallaby → Dalmatian

Institution	Deployment	Use Case
CERN (Geneva)	Rocky/Antelope, 10K+ VMs	LHC data processing, Grid computing
CINECA (Bologna)	Wallaby 2021.1 ADA Cloud	National HPC cloud for research (this course)
INFN (Italy)	Multiple deployments	Particle physics, HEP Grid
STFC (UK)	IRIS/JASMIN infrastructure	Climate data, genomics, astrophysics
OSG/XSEDE (USA)	OpenStack-based resources	Open Science Grid federated compute

OpenStack vs. AWS: Complementary, Not Competing

Same user-facing problem, opposite positions on the operational spectrum

Dimension	AWS	OpenStack
Type	Managed public cloud service	Open-source cloud platform software
Operator	Amazon	Your institution / service provider
Cost model	Pay-per-use (OpEx)	CapEx (hardware) + OpEx (operations)
Customization	Limited to available service catalog	Full control: every layer configurable
Data location	AWS regions (external)	Your datacenter
Operations burden	Managed by Amazon	Managed by your ops team
Feature velocity	Very high (thousands of engineers)	Slower (community-driven, 6-month cycle)
Multi-tenancy	Platform-enforced	Self-managed via projects/domains
HPC integration	With specialized instances	Full hardware flexibility
Compliance	Shared responsibility model	You control the full stack

Key insight: AWS maximizes convenience at the cost of control; OpenStack maximizes control at the cost of operational complexity.

OpenStack Architecture: Design Principles

Radical modularity + API-driven design: every operation is a REST call

1

Radical Modularity

COMPOSABILITY - Each functional domain (compute, network, storage, identity) is an independent service with a well-defined REST API. Deploy only what you need. Substitute alternative implementations.

2

API-Driven Design

Every operation (create VM, allocate floating IP, upload image) is a REST API call. Dashboard and CLI are both just API clients. Anything in the GUI can be automated with a script.

3

Asynchronous Operations

Services communicate via two channels: (1) REST API for synchronous queries/config; (2) AMQP message queue for async operations like VM provisioning, which may take seconds to minutes.

OpenStack Full Architecture

Layered service stack: REST APIs + message queue backbone

HORIZON : Web Dashboard (Horizon is optional; CLI/API is primary)

REST APIs: Each service exposes its own versioned endpoint (<https://ada.hpc.cineca.it:PORT/vX>)



KEYSTONE: Identity, Authentication, Authorization, Service Catalog (all services depend on Keystone)

Message Queue (RabbitMQ) • Database (MariaDB) • Shared State

Physical Infrastructure | Compute Nodes (KVM/QEMU) | Storage Backends (Ceph, LVM) | Network Fabric (OVS, VLAN)

Core Services: OpenStack ↔ AWS Mapping

Functional equivalents: OpenStack services are lower-level and more configurable

OpenStack Service	AWS Equivalent	Primary Function
Keystone	IAM + STS	Identity, authentication, authorization, service catalog
Nova	EC2	Compute: virtual machine lifecycle management
Neutron	VPC + Security Groups	Software-defined networking, virtual networks, routers
Cinder	EBS	Block storage: persistent volumes for VMs
Glance	AMI Registry (EC2)	VM image registry and metadata service
Swift	S3	Object storage: RESTful blob storage
Horizon	Management Console	Web-based dashboard (optional, CLI is primary)
Heat	CloudFormation	Infrastructure-as-Code orchestration

Mapping is functional, not structural. OpenStack services are lower-level: more configurable, but require more operational knowledge.

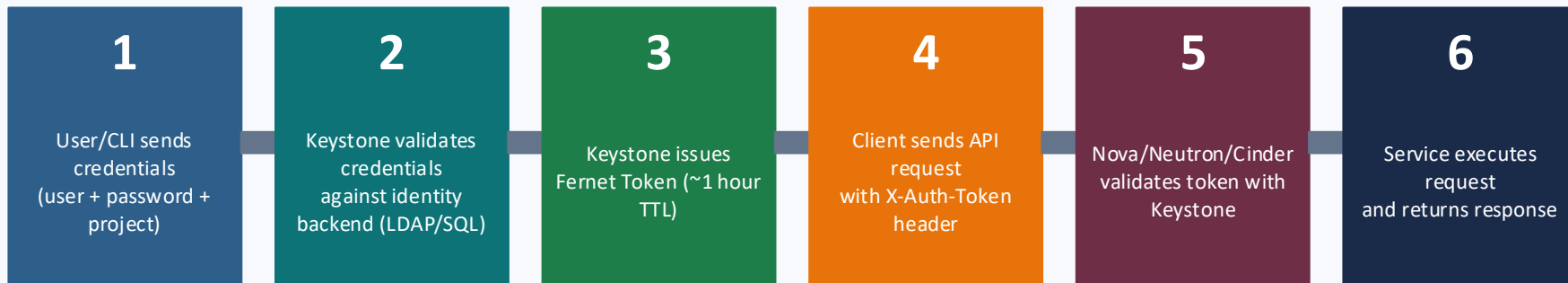
Keystone: Identity and Authentication Deep Dive

The central dependency: every OpenStack service validates tokens against Keystone

Domain	Top-level administrative boundary. Analogous to AWS Organization or IAM root account. CINECA uses one domain per institution.	Project	Resource and quota container within a domain. Analogous to AWS Account. All resources (VMs, volumes, networks) belong to a project.
User	Human or service principal with credentials. Users are assigned roles within projects. Service accounts use Application Credentials.	Role	Named permission set applied to a (user, project) pair. Common roles: admin, member, reader. Roles are local to the assignment scope.
Token	Short-lived bearer credential (~1h) issued after successful authentication. Passed as X-Auth-Token HTTP header to every API call.	Service Catalog	Directory of all available services and their endpoint URLs. Embedded in the token response. Tells clients where to send API requests.

Keystone Authentication Flow

Token-based model: how every OpenStack API call is authenticated



Application Credentials (Recommended for Automation)

Never embed username/password in scripts or openrc files committed to Git. Use Application Credentials: scoped to a project, revocable, can be set to expire. Created via: `openstack application credential create my-cred`

```
# Source credentials file (never commit to Git!)
source ~/openrc.sh

# Verify authentication
openstack token issue
openstack project list
```

CINECA ADA Cloud: Your Private Cloud Platform

OpenStack Wallaby deployment at Italy's national HPC centre

Platform: OpenStack Wallaby (2021.1) | URL: <https://adacloud.hpc.cineca.it> | Quota-based, no per-minute billing

71 nodes

Intel CascadeLake
8260 CPUs

48 cores

per node
(Hyperthreading)

768 GB

RAM per node
DDR4 ECC

2 TB

NVMe SSD
local storage

AWS Managed Cloud

Managed by AWS

Hardware

Managed by AWS

OpenStack Platform

Managed by AWS

Networking Fabric

YOUR RESPONSIBILITY

VMs / Containers

YOUR RESPONSIBILITY

Applications / Data

CINECA ADA (OpenStack)

Managed by CINECA

Managed by CINECA

Managed by CINECA

YOUR RESPONSIBILITY

YOUR RESPONSIBILITY

Authentication and CLI Setup

python-openstackclient — the unified CLI for all OpenStack services

Installation (pinned version for ADA)

```
python3 -m venv venv
source venv/bin/activate
pip install \
    python-openstackclient==5.8.0
```

openrc.sh — Credentials File

```
export OS_AUTH_URL=https://adacloud.hpc.cineca.it/identity
export OS_PROJECT_NAME=mhpc-2025
export OS_USERNAME=your-username
export OS_PASSWORD=your-password
export OS_REGION_NAME=ADA
```

Verification — Confirm you can authenticate and see your quota

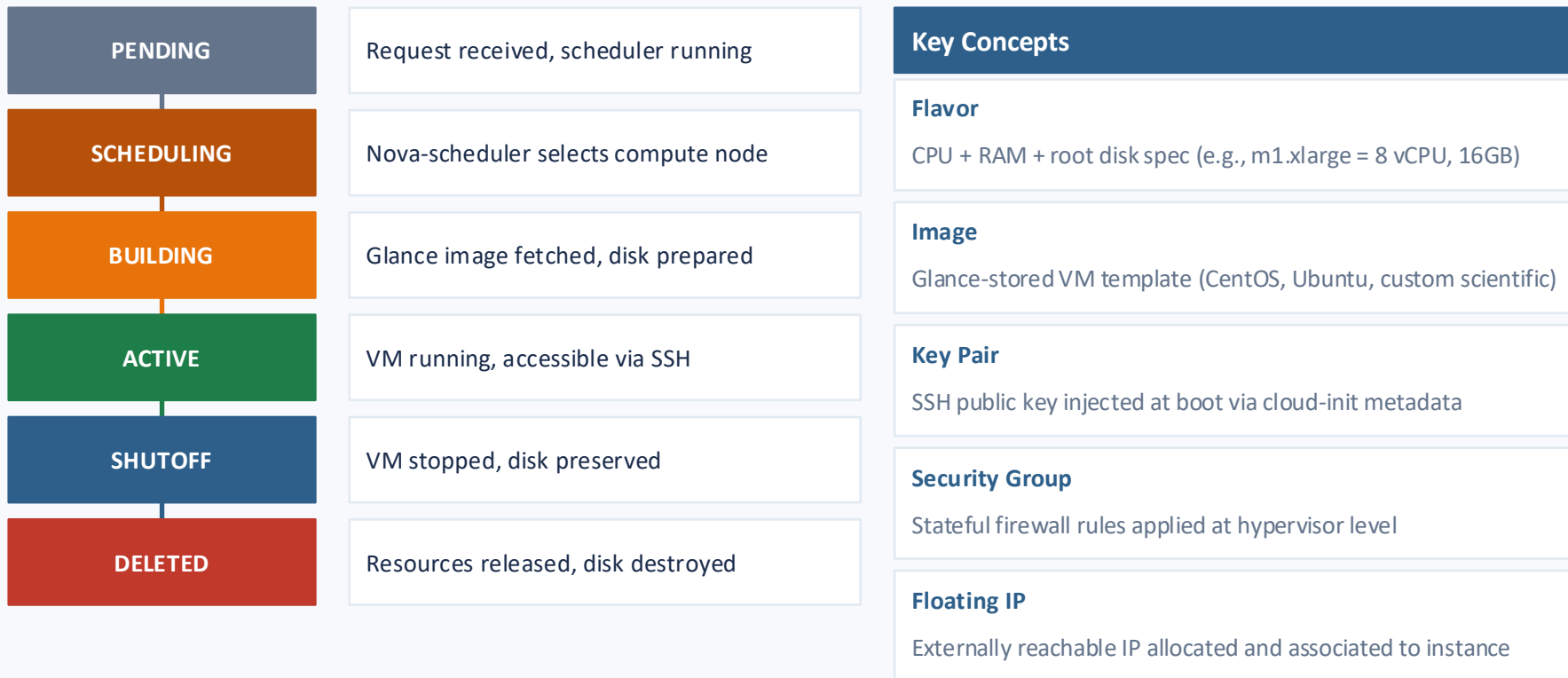
```
source ~/openrc.sh
openstack token issue           # Should return token with project scope
openstack quota show           # Inspect available resource limits
openstack flavor list          # Available VM sizes
```

Security Warning:

Never commit openrc.sh to Git. Use Application Credentials for automation (openstack application credential create). Store credentials in environment variables or a secrets manager, never in code.

Nova: Compute Instance Lifecycle

From flavor+image selection to running VM: what happens under the hood



Nova in Practice: Launching a Scientific VM

Complete CLI workflow: from listing resources to SSH access

```
# === Step 1: Discover available resources ===
openstack image list --community          # Available VM images (CentOS, Ubuntu...)
openstack flavor list                     # Available sizes (vCPU, RAM, disk)
openstack keypair list                    # Check your SSH keys
```

```
# === Step 2: Import your SSH public key ===
openstack keypair create --public-key ~/.ssh/id_rsa.pub mhpc-key
```

```
# === Step 3: Create security group with SSH access ===
openstack security group create my-sg --description 'MHPC research VM'
openstack security group rule create --proto tcp --dst-port 22 my-sg
```

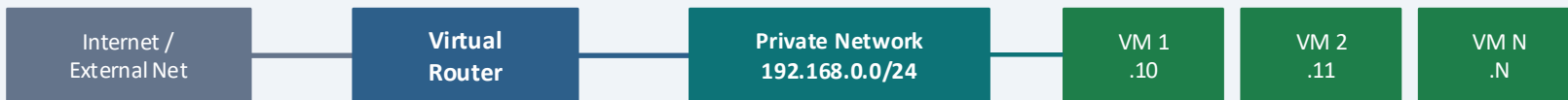
```
# === Step 4: Launch VM ===
openstack server create --image 'CentOS-8-stream' --flavor m1.xlarge \
  --key-name mhpc-key --security-group my-sg blast-analysis-vm
```

```
openstack server show blast-analysis-vm    # Check status → ACTIVE
```

```
# Step 5 - Floating IP: openstack floating ip create external && openstack server add floating ip blast-vm <FIP>
```

Neutron: Software-Defined Networking

Virtual networks, routers, floating IPs — AWS VPC equivalent



Floating IP

Externally routable IP that gets NAT'd to the private IP. Allocated from the external pool, then associated to a specific VM. Equivalent to AWS Elastic IP.

Security Groups

Stateful firewall rules at the hypervisor level. Applied per-instance (not per-subnet like AWS NACLs). Default: deny all inbound. Allow SSH (22), ICMP explicitly.

DNS

DNS resolution via external DNS or internal Neutron DNS. Instances get hostname in neutron.local.

```
# Create private network + subnet + router
openstack network create mhpc-net
openstack subnet create --network mhpc-net --subnet-range 192.168.10.0/24 --dns-nameserver 8.8.8.8 mhpc-subnet
openstack router create mhpc-router && openstack router set --external-gateway external mhpc-router
openstack router add subnet mhpc-router mhpc-subnet

# Allocate + associate floating IP
FIP=$(openstack floating ip create external -f value -c floating_ip_address)
openstack server add floating ip blast-analysis-vm $FIP
```

Cinder: Block Storage for Persistent Data

Persistent volumes that survive VM deletion: AWS EBS equivalent

Volume	Persistent block device, formatted with ext4/xfs. Survives VM deletion. Can be detached and reattached to another VM. Typical use: database, checkpoints.
Snapshot	Point-in-time copy of a volume. Used for backup and creating new volumes. Analogous to EBS snapshot → AML workflow.
Volume Type	Quality-of-service class (e.g., ceph-ssd vs ceph-hdd). Analogous to AWS EBS gp3 vs io2. ADA offers ceph-backed volumes.
Bootable Vol.	Volume from which a VM can boot. Persists the OS across VM termination (unlike ephemeral root disk). Essential for long-running research nodes.

Cinder CLI Workflow

```
# Create 100 GB research volume
openstack volume create \
  --size 100 --type ceph \
  blast-data-vol

# Attach to running VM
openstack server add volume \
  blast-analysis-vm blast-data-vol

# On the VM: format + mount
sudo mkfs.ext4 /dev/vdb
sudo mount /dev/vdb /data

# Snapshot for checkpoint
openstack volume snapshot create \
  --volume blast-data-vol \
  snap-pre-analysis
```

Part 1 — Key Takeaways

Private cloud

Same NIST cloud principles applied to self-owned infrastructure. Essential for data sovereignty, TCO at scale, HPC customization.

OpenStack

Open-source cloud OS: modular services, all API-driven, 12+ components. Dominant in European scientific computing (CERN, CINECA).

AWS ↔ OpenStack

Functional mapping: Nova=EC2, Neutron=VPC, Cinder=EBS, Glance=AMI, Swift=S3, Heat=CloudFormation, Keystone=IAM.

Keystone

Central dependency. Domain→Project→User→Role model. Every API call carries a Fernet token. Use Application Credentials for automation.

CINECA ADA

OpenStack Wallaby, 71 × Intel CascadeLake nodes. You manage VMs upward; CINECA manages the platform downward.

Networking+Storage

Neutron: private network → router → floating IP workflow. Cinder: persistent volumes survive VM deletion — always attach for research data.